MTH 200 **Integration & the FTOC**

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Determine the value of the definite integral $\int_1^e \left(\frac{x+x^2+1}{x}\right) dx$.

2. Why can't the definite integral $\int_0^e \left(\frac{x+x^2+1}{x}\right) dx$ be computed based on our current knowledge?

3. Consider the definite integral $\int_2^4 (x+1)\,dx$.

a) Compute the value of the integral using geometry.
b) Compute the value of the integral using the FTCP2.
c) Compute the value of integral using the limit definition of the definite integral.

4. Consider the definite integral $\int_0^2 (x^2-1)\,dx$.

a) Compute the value of the integral using the FTCP2.
b) Compute the value of integral using the limit definition of the definite integral.

5. Evaluate each of the following integrals.

a) $\int_0^1 (u+2)(u-3)\,du$

b) $\int_{\pi/2}^{\pi/4} \cot^2(x)\,dx$

c) $\int_{-2}^{1} (x^{-4}+1)\,dx$

d) $\int_0^1 (t^2+t+1)^2\,dt$

6. The function $H(x)$ is defined on $\left[0, \frac{\pi}{2}\right]$ by $H(x) = \int_{\cos(x)}^{\sin(x)} \sqrt{1-t^2}\, dt$.

a) Show that $H'(x) = 1$.

b) Compute the value of H at $\frac{\pi}{4}$.

c) Use the results of parts (a) and (b) to determine $H(0.2)$.

7. Compute the following limit $\lim_{h \to 0} \frac{1}{h} \int_{2}^{2+h} \sin(x^2)\, dx$. *Hint: Consider how rewriting this limit using properties of integrals can be helpful.*